Jie Liu

M.S. Student, Energy and Power Engineering, Shanghai Jiao Tong University
Research interests: stress corrosion cracking of nuclear engineering materials, additively manufactured stainless steels, high-temperature water environments
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Last updated:

Education

Shanghai Jiao Tong University, M.S. in Energy and Power Engineering Sep. 2024 - Expected Jun. 2027

GPA 3.53/4.0; First-Class Academic Scholarship. Research focuses on stress corrosion cracking of nuclear engineering materials in high-temperature water environments, combining microstructural characterization, crack statistics, data analysis, and finite-element simulation.

Shanghai Jiao Tong University, B.Eng. in Nuclear Engineering and Technology Sep. 2020 - Jun. 2024

GPA 3.33/4.3. Selected coursework: Nuclear Reactor Thermal Hydraulics, Two-Phase Flow and Heat Transfer, Nuclear Reactor Physics, Nuclear Reactor Safety Analysis, Numerical Methods for Reactor Engineering, Nuclear Fuel Cycle, Nuclear Environmental Engineering, Thermal System Design and Practice.

Research Output

- J. Liu, Y. Zhou, X. Shang, L. Zhang, K. Chen, Coupled mechanical and crystallographic analysis of intergranular stress corrosion cracking initiation in additively manufactured 316L stainless steels, *Corrosion Science* (2026) 113852.
- Y. Zhou, J. Liu, K. Chen, H. Shi, X. Shang, Z. Que, Z. Shen, L. Zhang, X. Zeng, Predicting intergranular stress corrosion cracking of stainless steels in high-temperature water by incorporating crystallographic factor, *Acta Materialia* 297 (2025) 121322.
- Y. Zhou, J. Liu, P.A. Ferreiros, X. Shang, K. Chen, Z. Que, Z. Shen, J. Yu, L. Zhang, Integrated effects of non-equilibrium microstructures on stress corrosion cracking susceptibility of post-treated laser powder-bed-fusion 316L stainless steels, *Corrosion Science* 252 (2025) 112974.
- Y. Zhou, H. Hu, C. Peng, J. Liu, Y. Zhao, X. Shang, C. Guo, Z. Shen, M. Song, L. Zhang, K. Chen, A precipitation hardened austenitic stainless steel with excellent stress corrosion cracking resistance against high-temperature water, *Corrosion Science* (2025) 113352.

Technical Skills

- **Programming and analysis**: Python, Matlab; experimental data processing, numerical computation, image-recognition workflows, and automated analysis.
- **Experiments and characterization**: electron microscopy, metallic microstructure analysis, stress corrosion cracking experiments, and crack statistics.
- **Simulation and modeling**: crystal plasticity finite-element simulation, thermal-system modeling, and SolidWorks 3D modeling.
- **Keywords**: stress corrosion cracking, additively manufactured 316L stainless steel, high-temperature water, crystallographic factors, machine learning.

Internship

Shanghai Nuclear Engineering Research and Design Institute, Engineering Equipment Department, Research Assistant Jun. 2023 - Jul. 2023

- Reviewed and reproduced calculation documents for Westinghouse passive residual heat removal heat exchangers; organized the tube outer-wall heat-transfer workflow and compared the influence of different heat-transfer correlations.

- Developed a Matlab-based PRHR HX heat-transfer performance calculator with a GUI for parameter input, calculation, and result display.

Projects

Stress Corrosion Cracking of 3D-Printed 316L Stainless Steel, Undergraduate Thesis Sep. 2023 - Jun. 2024

- Evaluated the influence of heat-treatment processes on stress corrosion cracking susceptibility of additively manufactured 316L stainless steel.
- Analyzed the relationship between microstructural features, crystallographic factors, and crack initiation behavior in high-temperature water environments.
- Built a crack-recognition workflow using OpenCV and Ultralytics YOLOv8 to improve automation and consistency in crack statistics.

Thermal System Design and Practice, Team Leader Mar. 2023 - Jun. 2023

- Designed and optimized a reactor secondary-loop thermal system, covering heat exchanger selection, loop configuration, and system-efficiency analysis.
- Coordinated team tasks, performed Matlab-based system modeling, and led the writing of major sections of the project report.
- Built and optimized a new loop design and applied a genetic algorithm to analyze key factors affecting cycle efficiency.

Stair-Climbing Robot and Jumping Robot, Design and Manufacturing Project Member Mar. 2022 - Jun. 2022; Sep. 2022 - Jan. 2023

- Researched motion schemes, performed motion analysis, and contributed to mechanical design for climbing and jumping mechanisms.
- Created SolidWorks 3D models and simulation animations for design presentation and structural validation.

Profile

Interdisciplinary background in nuclear engineering and material degradation, with experience spanning reactor thermal hydraulics, reactor physics, stress corrosion cracking, high-temperature water experiments, microstructural analysis, crack recognition, data analysis, and finite-element simulation. Interested in applying machine learning and image recognition to materials reliability evaluation, with strong self-learning ability, project ownership, and cross-disciplinary collaboration skills.